Analysis

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1 Helpful Lemmas & Inequalities

Lemma 1.
- Let $x, y \in \mathbb{R}$. If for all $\varepsilon > 0$ we have $x - y < \varepsilon$, then $x \leq y$.
- Let $x, y \in \mathbb{R}$. If for all $\varepsilon > 0$ we have $|x - y| < \varepsilon$, then $x = y$.

Proof: Note that if $x > y$ then $x - y \geq \frac{x-y}{2} = \varepsilon$, which proves the contrapositve. The second part followsw from the first since $|x - y| < \varepsilon \implies x - y < \varepsilon \wedge y - x < \varepsilon$.

2 Differentiation

Definition 2 (Compactness)**.** A set $S \subset U$ (in a topological space) is called **compact** if for every open cover $\mathcal{T}$ of S there is a finite subcover of S.

Theorem 3. The image of a compact set under a continuous function is compact.

Proof idea: If $f : D \to \mathbb{R}$ is a continuous function and D is compact, then let $\mathcal{T}$ be any open cover of $f(D)$. Then taking the preimage of each $T \in \mathcal{T}$ gives an open cover of D. This has a finite subcover, $\mathcal{T}_0$. Then mapping these back to elements of $\mathcal{T}$ gives a finite subcover of $f(D)$.

Theorem 4. Let S be a compact set on $\mathbb{R}$ and $f : S \to \mathbb{R}$ be continuous. Then f achieves both its maximum and minimum on S.

Proof: By 3 theorem, $f(S)$ is compact. By Heine-Borel, it is closed and bounded and hence $\sup f(S) = \max f(S)$ and likewise for the minimum. $\square$

Theorem 5 (Rolle's Theorem)**.** Let $f : \mathbb{R} \to \mathbb{R}$ be a function continuous on $[a, b]$ and differentiable on (a, b) such that $f(a) = f(b)$. Then there exists $c \in (a, b)$ s.t. $f'(c) = 0$.

Theorem 6 (Mean Value Theorem)**.** Let $f : \mathbb{R} \to \mathbb{R}$ be a function that is continuous on $[a, b]$ and differentiable on (a, b). Then there exists a $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Proof: Consider the function $g(x) = f(x) - \frac{f(b)-f(a)}{b-a}(x - a)$. Then $g(a) = f(a)$ and $g(b) = f(a)$. Thus by Rolle's Theorem there exists a $c \in (a, b)$ s.t. $g'(c) = 0$ and the rest follows. $\square$

Theorem 7 (Taylor's Theorem)**.** Let $f : \mathbb{R} \to \mathbb{R}$ be a function that has its first n derivatives continuous on $[a, b]$ and differentiable on (a, b). Fix a point to expand around, x_0. Let $x \in [a, b]$ with $x \neq x_0$. Then for any $n \in \mathbb{N}$ there exists and $c \in (x, x_0)$ such that

$$f(x) = \sum_{k=0}^{n} \frac{f^{(n)}(x_0)}{k!}(x - x_0)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1}$$

Idea of Proof:

- Construct a function $F(t)$ such that
  1. F is continuous on (a, b) and differentiable on $[a, b]$.
  2. $F(x) = f(x)$ and $F(x_0) = f(x)$
  3. $F'(t) = 0$ forces the correct coefficient of $(x - x_0)^{n+1}$
- Use the mean value theorem on $F(t)$.

Noteworthy property of Modified Taylor Series: If $P(t) = \sum_{k=0}^{n} \frac{f^{(k)}(t)}{k!}(x - t)^k$ then

$$P'(t) = \frac{f^{(n+1)}(t)}{n!}(x - t)^n$$

Proof:(following Lay 28.2) Let $x \in [a, b]$ and fix $n \in \mathbb{N}$. Let

$$F(t) = \sum_{k=0}^{n} \frac{f^{(k)}(t)}{k!}(x - t)^k + M(x - t)^{n+1}$$

where M is the unique solution to

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + M(x - x_0)^{n+1}.$$

Note that by the noteworthy property mentioned above we have

$$F'(t) = \frac{f^{(n+1)}(t)}{n!}(x - t)^n + (n + 1)M(x - t)^n.$$

Note also that $F(x) = F(x_0) = f(x)$, so invoking the mean value theorem (or Rolle's Theorem) we obtain that there exists a c between x and x_0 strictly such that

$$F'(t) = 0 = \frac{f^{(n+1)}(t)}{n!}(x - t)^n + (n + 1)M(x - t)^n.$$

Solving for M gives $M = \frac{f^{(n+1)}(x_0)}{(n+1)!}$ as required. □

Example 8. Let $f(x) = x^2 + 2x + 1$ on $[0, 3]$ with $x_0 = 0$. Let $n = 1$. Then there should exist a c such that

$$f(2) = 1 + 2(x - 0) + \frac{f^{(n+1)}(c)}{(n+1)!}(x - 0)^{n+1}$$

$$1 + 2x + \frac{2}{2}x^2$$

so we recover our polynomial.

We now note that we can rephrase this theorem in terms of a remainder.

Theorem 9 (Taylor's Remainder Theorem)**.** Let $f : \mathbb{R} \to \mathbb{R}$ be $n+1$ times differentiable and $x_0 \in \mathbb{R}$. Then if we approximate

$$f(x) \approx p_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^n$$

then their exists a c between x and x_0 such that

$$f(x) - p_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1}.$$

Note thought, that c depends on x.

Remark 10. Note that if f is a polynomial of degree k, then $p_n = f$ for all $n \geq k$ and all $x_0 \in \mathbb{R}$.

Lemma 11. Let f exist in a neighborhood of c and that $f''(c)$ exists. Then

$$f''(c) = \lim_{h \to 0} \frac{f(c+h) + f(c-h) - 2f(c)}{h^2}$$

$$f''(c) = \lim_{h \to 0} \frac{(f(c+h) - f(c)) + (f(c-h) - f(c))}{h^2}$$

Proof: Starting with the limit and using L'Hospital's rule we get:

$$\begin{aligned}
\lim_{h \to 0} \frac{f(c+h) + f(c-h) - 2f(c)}{h^2} &= \lim_{h \to 0} \frac{f'(c+h) - f'(c-h)}{2h} \\
&= \lim_{h \to 0} \frac{f'(c+h) - f'(c) + (f'(c) - f'(c-h))}{2h} \\
&= \frac{2f''(c)}{2} = f''(c)\square.
\end{aligned}$$

Lemma 12. The following are equivalent definitions of the derivative.

- The usual one
- $f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c}$
- $f'(c) = \lim_{h \to 0} \frac{f(c+h) - f(c-h)}{2h}$

3 Integration

Definition 13. A partition of $[a, b]$ is an ordered, finite subset P of $[a, b]$ indexed from 0 to n.

$$P = \{a = x_0, x_1, x_2, \ldots, x_n = b\}$$

Definition 14. For a partition P of $[a, b]$ define the **Upper Sum** of a bounded functino f to be

$$U(f, P, [a, b]) = \sum_{i=1}^{n} \Delta x_i \cdot \sup\{f(x) : x \in [x_{i-1}, x_i]\}$$

Similarly, the lower sum is

$$L(f, P, [a, b]) = \sum_{i=1}^{n} \Delta x_i \cdot \inf\{f(x) : x \in [x_{i-1}, x_i]\}$$

Definition 15. The Riemman Integral of f if it exists is

$$\int_a^b f(x)dx = \inf_{Q \geq P} U(f, Q, [a, b]) = \sup_{Q \geq P} L(f, Q, [a, b])$$

for any starting P (or just $\{a, b\}$) if the two numbers agree.

It may be interesting to note that the set of partitions is a sublattice of the lattice of sets with $\cup$ as a join and $\cap$ as a meet (or vice versa).

Lemma 16. Note that for any two partitions P, Q we have

$$L(f, P) \leq U(f, Q)$$

Proof: Consider the join $T = P \cup Q$, then by monotonicity of L and U under refinement we have

$$L(F, P) \leq L(f, T) \leq U(f, T) \leq U(f, Q)$$

. $\square$

Example 17. Note the Dirichlet function is not integrable by showing $U(f) = 1$ but $L(f) = 0$ over $[0, 1]$.

Example 18. Let's do an integral from the defition.

$$\int_0^b xdx = \frac{b^2}{2}$$

Let $P_n = \{k\frac{b}{n} \ : \ 0 \leq k \leq n\}$ be a partition for each $n \geq 1$. The we compute $U(x, P_n, [0, b])$ and L as well. Since the sup over any interval is always at the right hand side for $f(x) = x$ we have

$$U(x, P_n, [0, b]) = \sum_{k=1}^{n} k\frac{b}{n}\left(\frac{b}{n}\right) = \frac{b^2(n+1)}{2n}$$

$$L(x, P_n, [0, b]) = \sum_{k=0}^{n-1} \frac{kb}{n}\left(\frac{b}{n}\right) = \frac{b^2(n-1)}{2}$$

Note now that as $n \to \infty$ both L and U approach $\frac{b^2}{2}$.

Theorem 19. The following classes of functions are Riemann Integrable:

1. Monotone functions on $[a, b]$.

2. Continuous functins on $[a, b]$.

3. (Lebesgue-Vitali Thm) A bounded function on a compact interval $[a, b]$ is Riemann Integrable if and only iff it is continuous almost everywhere.

4. A bounded function on $[a, b]$ that has only a countable set of discontinuities, then it is Riemann Integrable.

Proof:
(1) Let $\varepsilon > 0$ and choose n such that $\frac{b-a}{n} f(b) < \varepsilon$. Then consider the uniform partition of $[a, b]$ into n parts, P_n.

$$U(f, P_n) - L(f, P_n) = \sum_{i=1}^{n} (f(x_i) - f(x_{i-1}))\Delta x$$

which telescopes to

$$f(b)\Delta x < \varepsilon.$$

and we are done.

(2) Recall that a continuous function on a compact interval is uniformly continuous, meaning for any $\varepsilon > 0$, there is a uniform $\delta > 0$ such that whenever $|x - y| < \delta$ we have $|f(x) - f(y)| < \varepsilon$. Observe that

$$U(f, P_n) - L(f, P_n) = \sum_{k=1}^{n} (\sup_{[x_{i-1}, x_i]} f - \inf_{[x_{i-1}, x_i]} f)\Delta x$$

so that choosing δ so that $|f(x) - f(y)| < \frac{\varepsilon}{b-a}$ and n large enough that $\Delta x < \delta$ gives

$$\sum_{i=1}^{n} (\sup_{[x_{i-1}, x_i]} f - \inf_{[x_{i-1}, x_i]} f)\Delta x < \sum_{i=1}^{n} \frac{\varepsilon}{b-a} \Delta x = \varepsilon$$

(3) We will come back to prove some other time.

(4) Follows from (3).

Theorem 20. Let f, g be integrable functions on $[a, b]$. Then

1. For any $k \in \mathbb{R}$ we have $\int_a^b kf = k \int_a^b f$.

2. $\int_a^b f + g = \int_a^b f + \int_a^b g$.

Theorem 21. If f is integrable on both $[a, b]$ and $[b, c]$, then f is integrable on $[a, c]$ and we have

$$\int_a^c f = \int_a^b f + \int_b^c f.$$

Theorem 22. If f is integrable on $[a, b]$, and g is continuous on a compact interval containing $f([a, b])$, then $g \circ f$ is integrable on $[a, b]$.

Proof Idea: Vaguely speaking, since g is uniformly continuous, and f is integrable, f should be continuous almost everywhere. Hence the spots that have potentially large variation occur on a small set, and the rest can be handled with the uniform continuity of g(ish). So choose a suitable partition P and split the subintervals into two categories. Those that $|f(x_{i-1}) - f(x_i)| < \delta$ and those that fail this. Show both sets have $U - L < \frac{\varepsilon}{2}$.

Corollary 23. If f is integrable on $[a, b]$, then so is $|f|$ and furthermore

$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$$